Today's Agenda

- Review for Midterm

Paper Reading Presentation Rubrics

Category	Description	Worth
An introduction to the problem and background	Did you make clear the nature of the problem you were trying to solve?	10
Methodology	Did you show a clear and deep understanding of the methodology developed in the paper?	30
Experiment	What were the datasets used for evaluation? How did the experiments setup	10
Critical Comments	Pros and Cons of the proposed methodology, potential issues to use the method	15
Visual Aids	Were visual aids used effectively? Were slides clear and easy to read by the audience?	10
Clarity & Organization	Was the presentation easy to understand; did it have a logical flow and organization?	10
Timing	Was the presentation well-paced? Did it fit within the time allotted?	10
Question	How well did you respond to questions?	5

Review of Chapter 2- Chapter 5

Chapter 2

- Human vision system
- Basics of image processing

Chapter 3

- Intensity transformation
- Spatial filtering

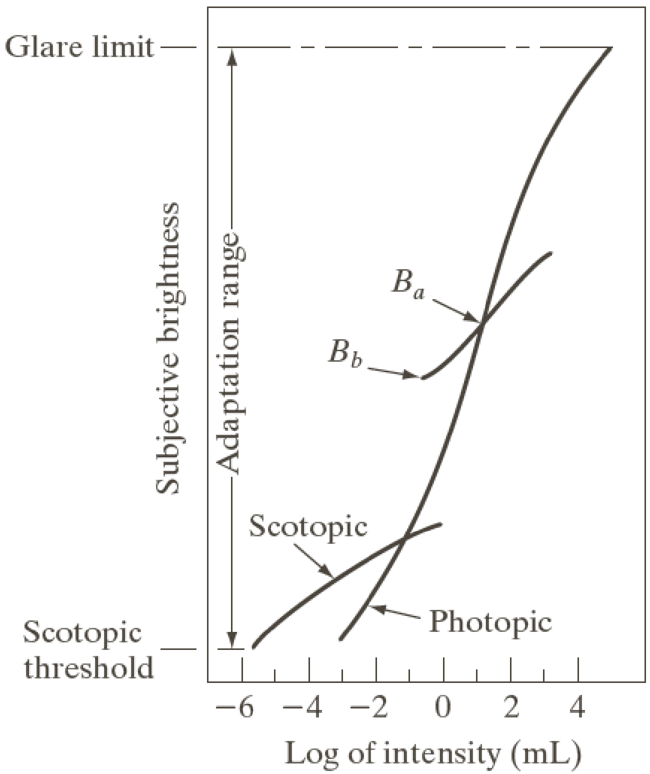
Chapter 4

- Fourier transform
- Image convolution in frequency domain

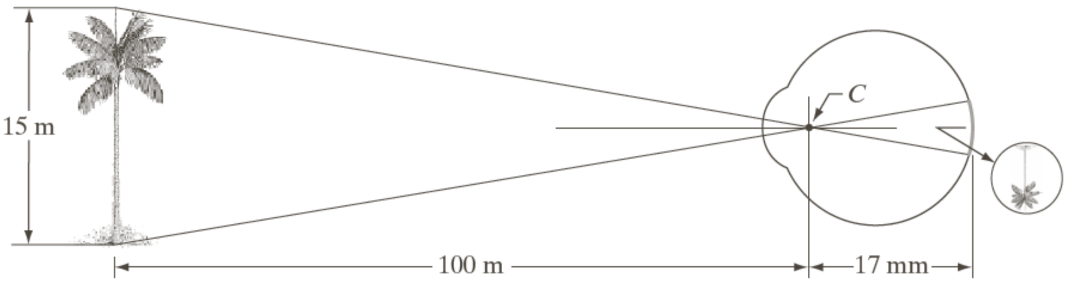
Chapter 5

- Image denoise
- Image degradation
- Image restoration

Human Vision System



Brightness adaption



- Geometrical relationship between the real object and the image of the object
- The minimum size of the object you can "see"

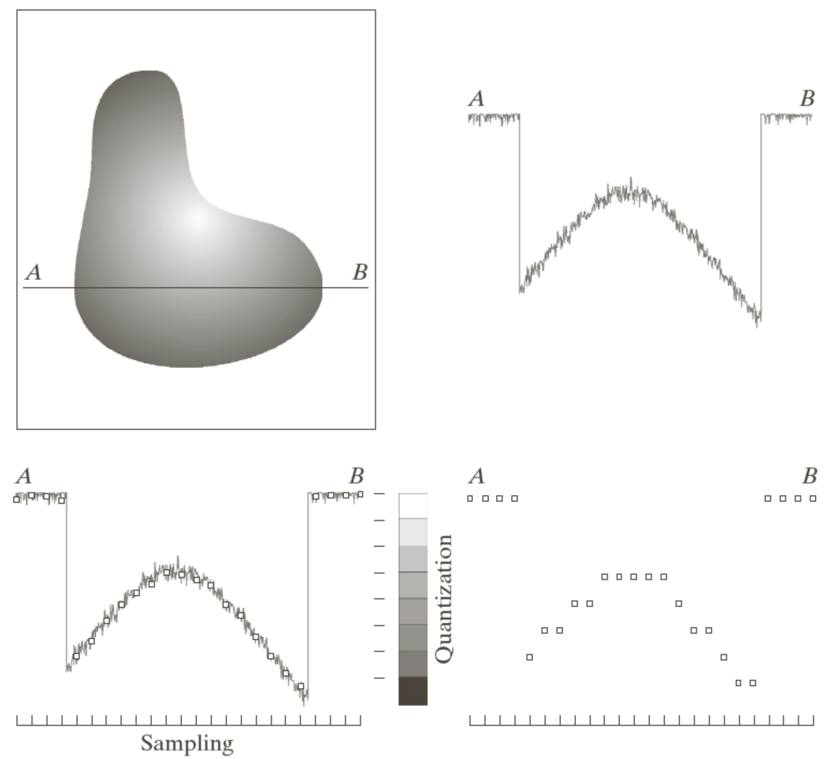
Basics of Image Processing

$$f(x,y) = i(x,y) \cdot r(x,y)$$

- **Image sampling and quantization**
  - Spatial/intensity resolution
- **Dynamic range of the image**

$$I_{\max} / I_{\min}$$

- **Image representation and storing**
- **Image interpolation**
  - Nearest neighbor and bilinear
- **Set operations**



Basics of Image Processing (Cont.)

Basic relationships between pixels

- Adjacency
- Connectivity
- Path

Basic relationships between regions

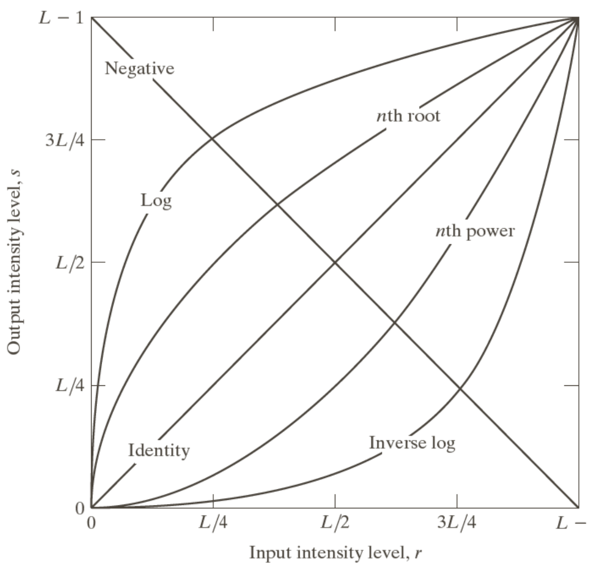
- Adjacency
- boundary

Distance measurement

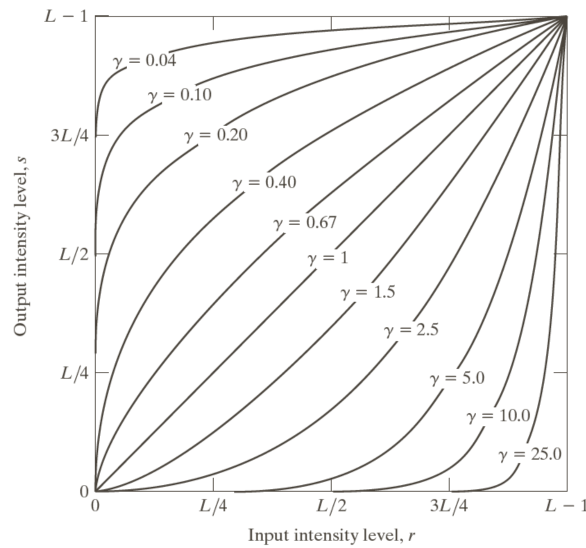
Mathematical tools

- Difference between matrix and array operation
- Linear/nonlinear operation
- Applications of image averaging, subtraction, and multiplication

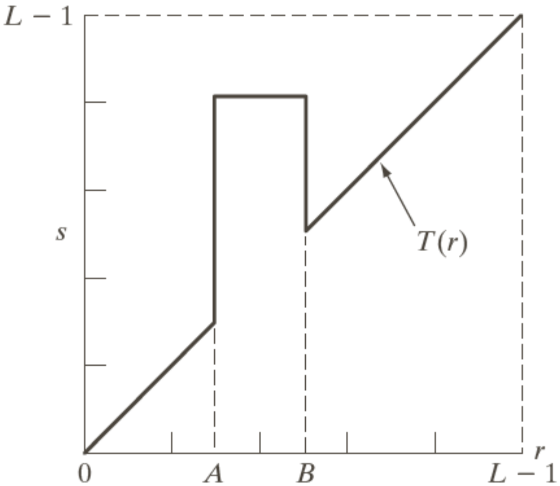
Intensity Transformation



Log transformation



Power-law (gamma) transformation



Intensity level slicing

Applications and working conditions using these transformations

Histogram Processing

What is a histogram of an image?

Histogram equalization

Histogram matching

Spatial Filtering

Image convolution in spatial domain and has properties of

- Commutativity, Associativity, distributivity

Image correlation in spatial domain

Spatial filters

- Smoothing filter
  - Average filter
- Sharpening filter
  - Laplacian filter
  - Unsharp masking
  - Sobel operator
- Order-statistic filter
  - Median filter
  - Min/max filter

Given a filter, you should know what it is use for

You should know what filter you can use for a specific task

Fourier Transform

Fourier series

$$f(t) = \sum_{n=-\infty}^{+\infty} c_n e^{\frac{j2\pi nt}{T}}$$

Unit impulse and its sifting property

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$$

Fourier transform

$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t}dt \Longleftrightarrow f(t) = \int_{-\infty}^{\infty} F(\mu)e^{j2\pi\mu t}d\mu$$

Image convolution in frequency domain

Basic Properties of FT

Linearity $h(t) = af(t) + bg(t) \leftrightarrow H(u) = aF(u) + bG(u)$

Translation $h(t) = f(t - t_0) \leftrightarrow H(u) = e^{-j2\pi t_0 u} F(u)$

Modulation $h(t) = e^{j2\pi u_0 t} f(t) \leftrightarrow H(u) = F(u - u_0)$

Scaling $h(t) = f(at) \leftrightarrow H(u) = \frac{1}{|a|} F(\frac{u}{a})$

Conjugation $h(t) = f^*(t) \leftrightarrow H(u) = F^*(-u)$

Symmetry $f(t) \leftrightarrow F(\mu) \Rightarrow F(t) \leftrightarrow f(-u)$

Image Degradation

$$g(x,y) = h(x,y) \otimes f(x,y) + \eta(x,y)$$

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

Important noise models

- Gaussian noise model
- Impulse noise model

Image denoise

- Various mean filters and their applications
- Order-statistic filters and their applications

Image restoration

- Inverse filtering, Wiener filtering, and Constrained Least Square filtering
  - Working conditions